CS452 HW2 - Buddy Memory System ReadMe

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1 Introduction

In this asssignment we were tasked with implementing a buddy memory system. A buddy memory system is a method of managing memory by repeatedly splitting it into halves when allocating memory and merging matching halves with freeing memory. In this methodology, memory blocks are allocated to always have sizes that are powers of 2: 8,16,32,...,1024,... bytes. In this implementation, we were to create a balloc function which is a stand in for malloc but with the desired buddy memory system functionality. One can make a call to balloc by calling bcreate(memory block size, lower bound, upper bound) where the memory block size is the size of the entire available memory for the buddy system, the lower bound is the smallest block allocateable, and the upper bound is the largest allocateable block. The way memory is allocated is that if a block of an arbitrary size is asked for, the allocator will continually divide the maximum memory block size in half until the size is $\geq$ to the desired size. So if 60 bytes was desired, a block of 64 bytes would be allocated. Of important note, when a block is split, two buddies of size 1/2 are created, hence this implementation being called the buddy memory system. The allocator then uses a freelist to keep track of which blocks are available it does this by taking advantage of a bitmap to reduce the memory overhead of saving this information.

2 File Contents

```
HW2/
├── GNUmakefile
├── ReadMe.pdf
├── deq.c
├── deq.h
├── error.h
├── main.c
├── main_2_balloctests.c
├── balloc.c
├── balloc.h
├── bbm.c
├── bbm.h
├── bm.c
├── bm.h
├── freelist.c
├── freelist.h
├── utils.c
├── utils.h
└── wrapper.c
```

3 Building and Running

To compile and run this program, navigate to the HW2 directory. Once inside, run the command:
$make
This will compile the program. To run the program using DEQ as the test file stay in the HW2 directory and run the command:
$./try
To run the tester file I specifically made to test balloc.c and freelist.c (main_2_balloctests.c), run the command:
$./balloctests
This file contains 53 tests to test the functionality and behavior of balloc and freelist.

4 Testing

Testing was divided into three sections: testing the freelist, testing balloc, and testing the double ended queue. The file main_2_balloctests.c contains the tests for freelist and balloc. In this file there are 10 tests for free list testing: creating memory, creating fthe free list itselft, checking the free list size, allocation testing, tests on empty listand unallocating memory. In the second section of main_2_balloctests.c I setup 43 tests to test the balloc functions. These tests can be split into 3 main areas: testing balloc itstelf, testing bsize, and exhaustion

testing. Finally main.c is the main file from the DEQ HW and is used to in conjuction with the wrapper.c file to check the functionality of replacing malloc with balloc. These tests are the same as for HW1, and check the complete functionaliry of a DEQ. All tests in all sections passed.